

PARAMPARA

Feel the Master's Hand

One sensor sleeve, from the fingers to the shoulder. Three living crafts: tabla, handloom and puppetry. We measure how a master moves, prove it is the master's, teach it, and step back until the learner performs alone.

Final solution document · version 4

AICTE · MIC Student Innovation · Heritage & Culture · Hardware

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In one minute

- **One innovation:** a master's Skill Fingerprint (how every finger, the wrist, elbow and shoulder move, plus what the tool does), used as a teaching reference.
- **One platform, three crafts:** tabla (rhythm), handloom (coordination and force), Kathputli puppetry (fine finger control).
- **Proven, not assumed:** the fingerprint must still name the master on a new day and a swapped instrument, or we drop the claim.
- **Learning counted with the device off,** and the master owns the recording.

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Citations like [12] point to Section 14. Every reference carries a status: **V** verified by live web search in October 2026, **P** existence verified with some details from memory, **S** a standard textbook or classic paper. Numbers we have not yet measured are marked **[Target]**.

1. The idea on one page

The problem

In tabla, handloom weaving and string puppetry, a master's skill lives in the hands and arms: which finger moves, how the wrist, elbow and shoulder work together, and the rhythm of it all. It passes on only face to face, over years. India's handloom workforce fell from 43.32 lakh to 35.22 lakh between the last two censuses [66]; Delhi's puppeteer colony was demolished in 2017 [73]; and archives keep how masters **sound and look** [50], not how they **move**.

Our one innovation

A master's Skill Fingerprint: a measured, consented record of how one master moves (every finger, the wrist, elbow and shoulder) and what their tool does, proven to belong to the master and not to their instrument or the day, and used as a teaching reference.

One sensor sleeve and one software pipeline serve three crafts. Everything else in PARAMPARA exists to capture, prove, teach or protect that fingerprint.

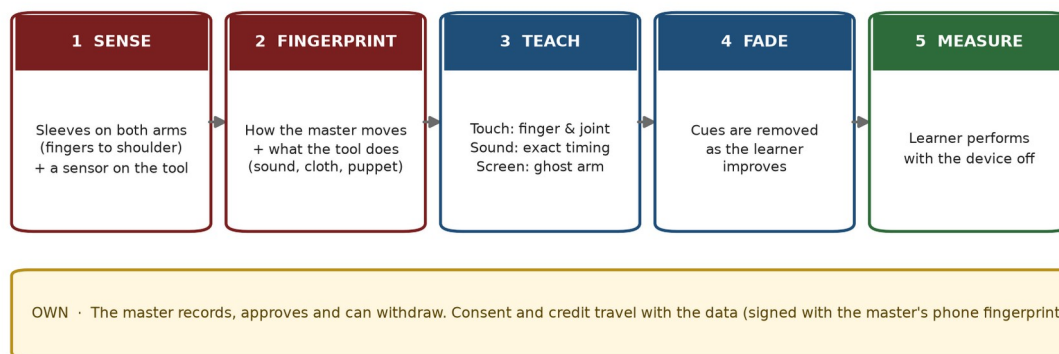


Figure 1. Five steps, plus ownership by the master.

Why three crafts, and why these three

Craft	Kind of skill	What the tool sensor measures	The 'cycle' we compare
Tabla	Fast striking rhythm	A piezo on each drum: stroke time and strength	One tal cycle (16 beats of Teentaal)
Handloom	Whole-arm coordination and force in a steady rhythm	Beater motion, treadle switches, a phone photo of the cloth	One pick: open the shed, throw, beat
String puppetry (Kathputli)	Fine, expressive finger control	A small motion sensor inside the puppet	One gesture phrase, e.g. a walk or a bow

A strike, a coordinated cycle and a fine expressive gesture are three different kinds of hand skill. **If one sleeve and one pipeline work for all three, they can serve most hand crafts.** For SIH, tabla is the lead craft and is validated in full; handloom and puppetry are validated on the two questions that matter most (Section 5).

Why not video, or a motion-capture suit?

Video shows the master but cannot measure the learner or check learning without help. Motion-capture suits measure movement but cost about US\$5,000 to over US\$12,000 [64, 65] and are built

for animation and VR, not for teaching one master's way. **PARAMPARA measures fingers to shoulder at a fraction of that cost, teaches one master's way, and proves the learner can do it alone.**

What the jury will see live

1. **The sleeve.** The judge wears it, plays a tabla phrase or moves a Kathputli puppet, and feels which finger and which joint to move.
2. **The proof.** Real masters' fingerprints, and the test showing they belong to the master, not to the instrument or the day (Section 5).
3. **Device off.** Cues fade, the judge performs alone, and the screen shows the unaided score.

Evidence status, stated plainly

Done today: a design grounded in published research (Section 7); open analysis code for the fingerprint, the confound test and the fade rule; the confound test shown on synthetic data to accept a real fingerprint and reject a fake one (Figure 3); protocols, consent forms and pass/fail rules written **before** any data.

Not yet done: recordings of real masters, results of E0–E5, signed partner agreements. **We claim none of these until they exist.** Result templates are ready (Appendices B and C), so the finale deck shows the real numbers, whichever way they go.

Our test of success: *the device succeeds only when the learner no longer needs it.*

2. The problem: three crafts, one gap

2.1 Skill that cannot be written down

All three crafts are learned by sitting with a master and imitating for years. Michael Polanyi called this tacit knowledge: 'we know more than we can tell' [18]; turning it into a shareable form is the hardest step in knowledge transfer [19]. Notation records the bols, a draft records the weave pattern, a script records the story. **None records the hands.**

2.2 The three crafts today

Craft	Situation	Source
Tabla	Taught face to face (<i>talim</i>). A school subject (CBSE 036) and a Kala Utsav category in 36 States/UTs. Ustad Zakir Hussain died in December 2024: his recordings stay, his teaching goes.	[32, 53, 54]
Handloom	35.22 lakh weavers and allied workers in 2019–20, down from 43.32 lakh. Training runs through 28 Weavers' Service Centres; 56,934 weavers trained under SAMARTH in six years. The national two-treadle qualification is 315 hours of hands-on practice. Handloom weavers report shoulder pain far more often than powerloom weavers (76% vs 42%).	[66, 67, 68, 69]
Puppetry	Four families of puppetry: string, shadow, rod and glove. In Kathputli the strings are looped on the puppeteer's fingers, with no control bar: the skill is literally in the fingers. Puppeteers face a shrinking audience; Delhi's Kathputli Colony (about 2,800 families of folk artists) was moved to a transit camp in 2017.	[23, 71, 72, 73]

Craft	Situation	Source
Policy	The State pays gurus to teach in person (Guru-Shishya Parampara), runs a scheme for intangible heritage, and counts about 64.66 lakh handloom and handicraft artisans. India has 16 elements on UNESCO's intangible heritage list.	[25, 26, 27, 29]

2.3 Why existing tools fall short

Tool	What it does well	What it cannot do
Video lessons and apps	Show the master's hands [33]	Measure the learner; check learning without help
Training centres and Guru-Shishya Parampara	The gold standard: a master in the room [27, 67]	Reach beyond one room, one master's time and one lifetime
Motion-capture suits	Measure body movement [64, 65]	Teach a specific master's way at an affordable cost; prove unaided learning
European craft-capture projects	Capture and represent craft gestures for museums and education [74, 75]	Teach one master's fingerprint through a wearable that fades out

3. The innovation: a master's Skill Fingerprint, fingers to shoulder

3.1 What the fingerprint is

Layer	What it captures	Measured by
Body	How each finger, the wrist, elbow and shoulder move in each cycle: which joint leads, when, how fast and how far	9 motion sensors per arm
Tool	What the movement produces: drum stroke time and strength; beater timing and force, cloth density; the puppet's motion	One tool sensor per craft
Rhythm	Where in the cycle the master is slightly early or late, and how steady	Body and tool sensors together
Consistency	How much the master varies from cycle to cycle	Spread over at least 20 cycles

The fingerprint is the average and spread of these features over at least 20 cycles. **Only features that pass the confound test (Section 5) are used for teaching.**

3.2 Why the whole arm, not only the hand

Published finding	Meaning for us	Source
Expert pianists move shoulder, then elbow, then wrist in a clear sequence; novices do not	Expertise shows in how the joints are ordered, not only in the fingers	[56]
Experts let the arm's own motion do work at the elbow and wrist, so their muscles do less	Efficient technique is visible in the whole arm	[57]

Published finding	Meaning for us	Source
Four drummers played the same accent with individual movement strategies	Movement carries personal style	[58]
Vibration at the joints in error cut real-time error by up to 27% and sped learning up to 23%	Joint-level touch cues work, but that study needed an optical lab	[59]
Kathputli strings are tied to the fingers	For puppetry, finger sensing is essential	[71]
76% of handloom weavers in a Varanasi survey reported shoulder pain	Measuring the shoulder also matters for weavers' health	[69]

3.3 Why we believe a fingerprint exists

Published finding	Meaning for us	Source
Tabla gharanas recognised automatically from audio of solo performances (38+ hours)	Style leaves a measurable trace	[39]
A large Hindustani corpus shows systematic accent and tempo patterns within the tal cycle	Accent and timing shapes are measurable	[38]
Pianists identified from timing and loudness; famous drummers each have a timing fingerprint	Performers can be told apart	[40, 41, 42]

None of these studies separated the performer from their instrument and recording session in a craft setting. That is exactly what our experiment E1 does (Section 5).

3.4 What is new

Closest existing work	What it does	What it does not do
TIKL vibrotactile suit [59]	Joint-level vibration to correct arm poses	Use a low-cost wearable (it used an optical lab); capture fingers; teach a specific master; fade; test device-off
Teslasuit, Xsens suits [64, 65]	Body motion capture (and haptics)	Teach; prove the learner can perform alone; cost suits a school or weaving centre
Mingei, CRAEFT [22, 74, 75]	Capture and represent craft gestures, simulators, haptics	A wearable that teaches one master's fingerprint and withdraws itself
Pottery gesture capture, i-Treasures [20, 21]	Recognise expert gestures with body sensors	Teach them back with fading and device-off measurement
Haptic Drum Kit and bracelets [4, 5]	Vibration cues for which limb plays	Finger-level cues; a specific master's style; retention
Open-palm data glove [60]	Accurate finger tracking with the palm free	Teaching or a master reference
Tabla gharana recognition [39]	Identifies style from audio	Body movement; teaching

Novelty statement

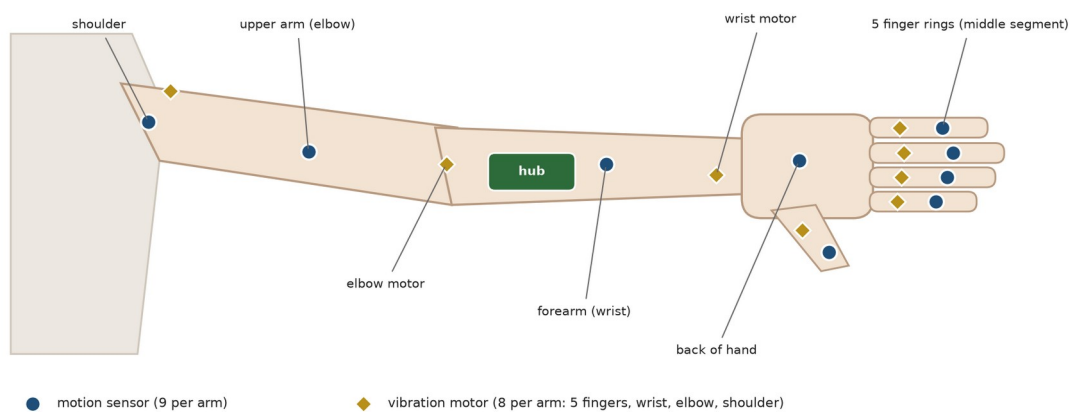
"To our knowledge, PARAMPARA is the first low-cost wearable that captures an individual master's skill fingerprint from fingers to shoulder together with what the tool does, proves that the fingerprint belongs to the master rather than to the instrument or the session, teaches it back through finger- and joint-level touch with sound and a ghost arm, withdraws the help as the learner improves, and counts only device-off performance, under the master's consent, across three Indian crafts on one platform."

3.5 If a fingerprint turns out to belong to the instrument

E1 decides this separately for each craft. **If a craft's fingerprint fails the confound test, we drop the style claim for that craft.** PARAMPARA then still teaches structure, rhythm and posture, measures device-off learning and archives consented recordings. We say this in advance because a claim we cannot test is not worth making.

4. How it works: Sense, Fingerprint, Teach, Fade, Measure, Own

4.1 Sense: the sleeve



Fingertips and palm stay free: sensors sit on the back of the fingers and arm, so the touch on drum, threads and strings is unchanged (checked in E0).

Figure 2. One sleeve per arm: 9 motion sensors and 8 vibration motors, from the fingers to the shoulder.

Part	Specification
Motion sensors (9 per arm)	Top of the shoulder, upper arm, forearm, back of the hand, and a ring on the middle segment of each finger and the thumb. 6-axis sensors (e.g. BMI270) at 100–200 samples per second; orientation from the Madgwick filter [61].
Vibration motors (8 per arm)	Base of each finger and thumb, wrist, elbow and shoulder. Each motor has its own driver chip (DRV2605L) for crisp, repeatable pulses [36].
Hub	ESP32-S3 on the forearm; two I2C multiplexers; rechargeable battery for 3–4 hours [Target]. Radio to a phone or laptop; timestamps shared with the tool sensors [37, 51].
Fingertips and palm	Left free. Nothing sits between the skin and the drum, the thread or the string. Open-palm designs still track fingers accurately [60].
Calibration	A 20-second standing pose (N-pose) at the start of each session [63].
Fit and hygiene	Washable fabric sleeve; rings in three sizes; sensors and motors clip out.

Accuracy, honestly. A single sensor's orientation is accurate to about a degree in the lab [61], but joint angles from body-worn sensors can be off by 10–20° at the shoulder, depending on the system and calibration [62]. So the fingerprint relies mainly on the **timing, order and speed** of the joints, which tolerate small placement errors, and uses angles only as ranges. E1 re-fits the sleeve on each recording day, so placement error is **tested, not assumed away**.

4.2 Sense: the tool

Craft	Tool sensor	Measures	Fitting
Tabla	One piezo per drum and a small base unit	Stroke time (under a millisecond), strength, which drum [17]	Clipped on with removable putty
Handloom	Motion sensor on the beater; switches under the treadles; phone photo of the cloth	Beat timing and force (peak acceleration); treadle order; picks per cm and their evenness [70]	Strapped on; nothing on the warp or the shuttle path
Puppetry	Small motion sensor and battery inside the puppet's head or body	How the puppet turns, tilts and moves	Inside the costume; strings untouched

4.3 Fingerprint

The tool sensor marks each cycle: a tal cycle on tabla, a pick on the loom, a gesture phrase on the puppet. For every cycle the software removes the cycle's overall tempo and size, then stores, for each joint, **when it moves relative to the tool event, which joint leads, how fast and how far**, together with the tool features and the rhythm. Averaging over cycles gives the fingerprint; the spread shows the master's consistency. **The code is the same for all three crafts; only the cycle definition changes.**

4.4 Teach: each sense does what it is best at

Sense	What it carries	Why
Touch: finger and joint motors	Which finger moves next; which joint should lead or is lagging; how strong the accent is	Joint-level vibration improved arm learning [59]; drum and strength cues were recognised 96% of the time [7]
Hearing	Exact timing: the master's tiny early and late shifts	The ear notices about 10 ms [43]; the skin needs 10–30 ms or more [46]
Sight: ghost arm	The master's arm as a moving figure over the learner's; replay after each cycle	Visual feedback suits movement shape and review between attempts [10]

Craft	Finger cue	Joint cue	Sound	Screen
Tabla	Which finger strikes (e.g. index for <i>na</i>)	Wrist lift before an accented beat	Bols at the master's timing	Ghost arms; accent profile
Handloom	Grip and release of the shuttle hand	Shoulder-then-elbow order of the beat; alert if the shoulder stays raised	The master's beat rhythm	Ghost arm; cloth evenness map
Puppetry	Which string-finger to pull, and how much	Wrist turn for the puppet's turn	Music cue	Ghost hand and ghost puppet

Timing the cue. Touch is dulled on a limb that is about to move [44, 45]. So each cue ends just before the movement starts and is set well above the learner's own sensing threshold.

Level	What the learner learns	Moves on when
1. Structure	The pattern: which finger, which joint, which stroke or step	Performs it without cues, 80% correct
2. Steady rhythm	Keeping the cycle steady while cues fade out	Unaided score $\geq 85\%$ on two different days
3. Style	This master's own timing, accents and joint order	Matches the master's fingerprint, and the master approves (only where E1 passes)

4.5 Fade

Help that never goes away makes learners dependent. Feedback given less often leads to better long-term learning, even though practice feels harder [13, 14]. The fade rule:

1. At the start of each session, and every four cycles, the learner performs one cycle with **no cues** (a check cycle).
2. The better the check cycles, the fewer cues the next cycles get.
3. If the learner struggles twice in a row, cues come back a little.
4. When the learner reaches the target without cues on two different days, the master is asked to approve the next level.

4.6 Measure: with the device off

Learning means what the learner can do **after a break and without help**, not how well they do while guided [12]. Each craft has an unaided score taken from check cycles and from a retention test at least a day later:

Craft	Unaided score (device silent)
Tabla	Timing, accent and stroke accuracy against the master's reference; steadiness and early/late tendency reported separately, because everyone plays slightly ahead of a beat [15]
Handloom	Evenness of picks per cm in the woven cloth (phone photo), steadiness of the beat rhythm, correct treadle–throw–beat order; time with the shoulder raised reported for health
Puppetry	How closely the puppet's motion follows the master's (after time alignment), and whether viewers who do not know the answer pick it as the master's

4.7 Own: consent and credit stay with the master

- The master decides how the recording may be used (teaching, sharing, commercial use, time limit) and can withdraw it.
- The master approves each recording, and each learner's progress, with the fingerprint sensor on their own phone. The secret key never leaves the phone; this is the passkey standard used for secure logins [49].
- Credit and the master's terms travel with the data as machine-readable labels [48]. An edited copy fails verification.
- This follows UNESCO's ethical principles for intangible heritage (free, prior and informed consent) and the CARE principles for community data [24, 47].

5. Validation: answering the hardest question first

5.1 The hardest question

"Are you identifying the master, or their instrument, tuning, recording session or sensor placement?"

This is a known trap. Music systems that appeared to recognise artists were partly recognising each album's recording conditions, the 'album effect' [76]. A naive test could fool us in the same way: if each master plays only their own tabla, loom or puppet on one day, a computer can 'identify the master' by recognising the instrument. **So E1 is designed to break that link.**

5.2 E1: a fingerprint that survives a new day and a new instrument

Recording plan per craft: 3 masters × 2 sessions on different days (sleeve taken off and re-fitted) × 2 instruments (masters swap tablas, looms or puppets) × 20 cycles = **240 cycles**.

Test	Question	Pass [decided in advance]	Rules out
T-a New day	Trained on one day, does it name the master on the other day?	≥ 70% (chance 33%)	Session, sensor placement
T-b New instrument	Trained on one instrument, does it name the master on the other?	≥ 70% (chance 33%)	Instrument, tuning
T-c Direct test	Is the same master on different instruments more alike than different masters on the same instrument?	Yes, with $p < 0.01$ (1,000 label shuffles)	All of the above together
Control	How well do the same features name the instrument?	Reported for transparency	Shows how much instrument signal remains

Verdict: PASS only if T-a, T-b and T-c all pass. Otherwise the style claim is dropped for that craft (Section 3.5).

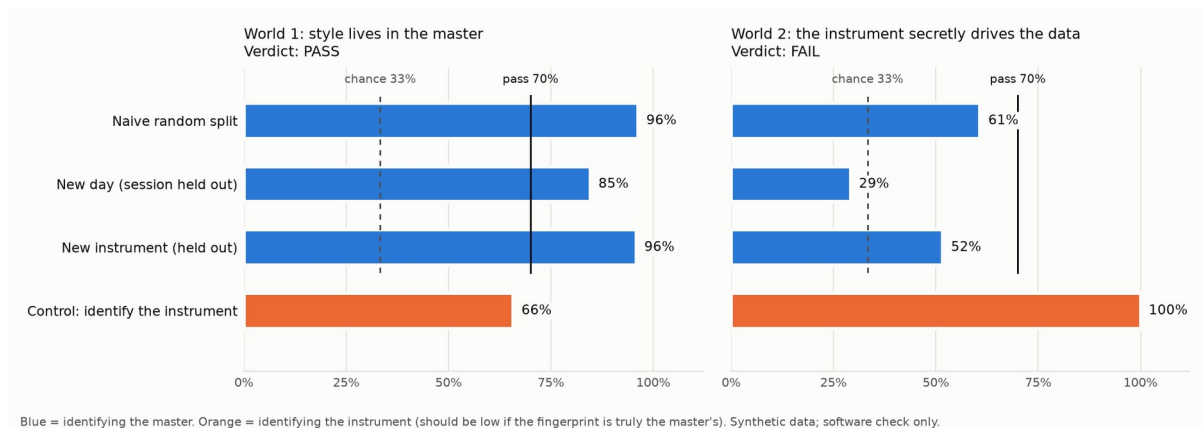


Figure 3. Software check on synthetic data. World 1: style lives in the master, and every test passes. World 2: the instrument secretly drives the data. A naive random split still looks well above chance (61%), but the new-day and new-instrument tests fall to 29% and 52%, and the verdict is FAIL. The test rejects a fake fingerprint, as it should. Instrument chance level is 50%.

```
python3 PARAMPARA/sim/skilltwin_validation.py demo # reproduces Figure 3
python3 PARAMPARA/sim/skilltwin_validation.py analyze e1.csv # real E1 data
```

5.3 All experiments, with pass rules fixed before data

#	Question	Design	People	Pass [Target]	Crafts at SIH
E0	Does the sleeve change how masters perform?	Alternating blocks of 10 cycles with and without the sleeve; the tool sensor measures both	Each master	Equivalent within ± 10 ms timing, $\pm 5\%$ cycle time, ± 1 dB strength (two one-sided tests [77]); comfort ≥ 4 of 5	All three
E1	Is the fingerprint the master's?	Section 5.2	3 masters per craft	T-a and T-b $\geq 70\%$; T-c $p < 0.01$	All three
E2	Can cues be felt while performing?	Weakest felt cue at rest vs while performing; finger, wrist, elbow and shoulder sites	8	Needs at most twice the rest strength, for at least one site per joint	Tabla, puppetry
E3	Can people tell two masters apart through PARAMPARA?	Feel and hear A, B, then X: is X like A or B? 20 rounds	10	≥ 15 of 20 right ($p = 0.02$)	Tabla
E4	Can cues be read while performing?	40 random cues: 5 fingers and 3 joints, 2 strengths	8–10	$\geq 90\%$ of sites, $\geq 80\%$ of strengths	Tabla, puppetry
E5	Do learners improve with the device off?	16 beginners randomised: video practice or PARAMPARA; 3 sessions on 3 days; retention test 48 hours later, device off	16	Report the difference in unaided score with a 95% confidence interval; run the full trial only if PARAMPARA is at least equal to video	Tabla

E5 is a **pilot**: 16 people cannot prove a small effect. It estimates the effect size and tests the procedure for the 120-person trial (Section 13), and we publish the result whichever way it goes. All studies are short, seated, low-risk tasks with written consent and institute ethics approval; pass rules are registered before recording.

6. What we build for SIH

6.1 Three deliverables

Deliverable	What the judge experiences	Done when [Target]
1. Two sleeves and three tool kits	Wears a sleeve, plays a tabla phrase or moves a puppet, and feels finger and joint cues	Cue timing varies ≤ 3 ms; 9 in 10 cues recognised while performing (E4)
2. Proven fingerprints	Sees real masters' fingerprints and the confound-test verdict for each craft	E0 and E1 reported for all three crafts, pass or fail
3. Device-off learning	Practises; cues fade; performs alone; sees the unaided score, and the E5 pilot result	Fade rule runs end to end; E5 reported with its confidence interval

6.2 Scope per craft

Craft	Validated at SIH	Live at the finale
Tabla (lead)	E0, E1, E2, E3, E4, E5	Judge plays with the sleeve; two masters' fingerprints; device-off score
Puppetry	E0, E1, E2, E4	Judge moves a Kathputli puppet and feels the string-finger cues
Handloom	E0, E1	Master's recording replayed on the ghost arm; cloth evenness from a phone photo; a tabletop loom if the venue allows

6.3 Bill of materials

Part	Qty	Approx. ₹
ESP32-S3 hub boards	2	1,400–1,800
6-axis motion sensor modules (9 per sleeve) [52]	18	4,500–7,200
Vibration motors (coin LRA)	16	3,200–6,900
Haptic driver boards (DRV2605L) [36]	16	4,800–7,200
I2C multiplexers	4	600–1,000
Batteries with chargers	2	800–1,200
Fabric sleeves, 3D-printed finger rings, cables, enclosures	2	1,600–2,400
Tool kits: tabla base unit with piezos; loom beater sensor and treadle switches; puppet sensor	3	2,500–3,700
Total prototype (two sleeves, three tool kits)		≈ 19,000–31,000

Estimates from retail module prices. At volume the chips cost about US\$1.26 (motion sensor) and US\$0.72 (driver) each [52], so a custom flexible circuit could bring one sleeve to about **₹5,000–7,000 [Target]**, against US\$5,000 or more for a motion-capture or haptic suit [64, 65].

Deliberately left out of the SIH build

Custom circuit board, on-device AI recognition, the Style level for any craft whose E1 fails, live remote teaching, and further crafts. These wait for the roadmap (Section 13).

7. Evidence from published research

7.1 Touch, sound and learning

Finding	Key number	Source
Joint-level vibration suit while copying a teacher's arm movement	Error down up to 27%; learning up to 23% faster	[59]
Adding haptic guidance to audio training in a drumming task	–17% final loudness error; –18% early timing error	[1]

Finding	Key number	Source
Haptic vs visual training of a movement	Timing learned better from haptics	[3]
Beginners learned intricate drum patterns from vibration cues alone	Haptic Drum Kit study	[4]
Violin bowing with vibration feedback	Improved; half kept the gain without feedback	[6]
Feeling a drum and strength cue on the body	96.18% recognised	[7]
Piano: vibration vs visual cues (n = 14)	Timing error 12.1% vs 22.3%	[8]
Feedback on half the trials vs every trial	Worse in practice, better retention	[14]

7.2 Sensing

Finding	Key number	Source
Low-cost orientation filter for wearable sensors	Below 0.8° static, 1.7° dynamic	[61]
Open-palm glove, fingertips free	Finger error 1.25°, wrist 4.85°	[60]
Body-worn sensors vs optical capture at the shoulder	About 10–20° in one system; under 4° for most joints in another	[62]
Cloth density measured from images	Error below 0.86% vs manual counting	[70]

7.3 What we will not claim

- That PARAMPARA transmits a master's 'soul'. We measure and teach **observable** parts of their skill.
- That it replaces the master. The master records, approves and controls.
- That a 16-person pilot proves long-term learning, or that the sleeve reduces weavers' pain. Both need larger studies.

8. Feasibility

8.1 Eight-week plan

Week	Build and record	Proof produced
1	Sleeve v1 on one arm: sensors, hub, logging, N-pose. Partner outreach in all three crafts (Appendix A). Ethics application; pass rules registered.	Sensor timing checked against slow-motion video
2	Second sleeve; motors and cue timing; three tool kits	E2 and E4 run on volunteers
3	Fingerprint pipeline and ghost arm; tabla masters, day 1	E0 (tabla); E1 day 1
4	Tabla masters, day 2 (instruments swapped, sleeve re-fitted); puppeteers, day 1	E1 verdict (tabla); E0 (puppetry)
5	Weavers at a weaving centre, days 1 and 2; puppeteers, day 2	E0 and E1 (handloom, puppetry); E3
6–7	Learning pilot: 16 beginners, 3 sessions each, retention test	E5 result with confidence interval

Week	Build and record	Proof produced
8	Full demo; three rehearsals without help; backup video; deck with real numbers	Demo runs end to end

8.2 Team of six

Role	Owns
Hardware	Sleeves, rings, tool kits, power
Firmware	Sensor reading, cue timing, radio, time sync
App	Ghost arm, lessons, scores, consent screens
Analysis	Fingerprint features, confound test, fade rule, E0–E5 statistics
Research and partner liaison	Outreach to masters, consent, recording sessions
Pitch	Deck, demo script, backup video

8.3 Top risks

Risk	What we do	Fallback
No masters available in time	Outreach in week 1 through Zonal Cultural Centres, Weavers' Service Centres, music colleges and puppeteer communities	Senior practitioners of different teachers, stated clearly
Masters cannot swap looms	Record two looms at one weaving centre	Report T-a only for handloom and say so
Sleeve changes how masters perform (E0)	Lighter rings; adjust positions with the master	Hand, wrist and elbow sensors only
Shoulder angles inaccurate	Use timing, order and speed features	Drop shoulder angles from the fingerprint
Finger cues not felt while striking (E2)	Cue before the movement; stronger pulse	Move the cue to the wrist
Fingerprint fails E1 for a craft	Report it	Teach structure and rhythm for that craft (Section 3.5)
Radio trouble at the venue	Cues stored on the sleeve in advance	USB cable; backup video

Every part is off the shelf, the riskiest questions are answered by week 5, and each risk has a fallback. **The SIH build is feasible in eight weeks; the results decide what we claim.**

9. Viability: institution-first, craft by craft

Our first customers are institutions, because hardware cost and the need to protect masters' content make direct consumer sales premature.

Craft	What already exists	What PARAMPARA adds
Tabla	Guru-Shishya Parampara honoraria (e.g. ₹7,500 a month for a guru) [27]; CBSE subject 036 [53]; ABGMVM exam centres [30]	A kit per guru or music room; fingerprints as lasting output of state-funded teaching
Handloom	28 Weavers' Service Centres and SAMARTH training [67]; a 315-hour NSQF qualification [68]	A training aid with device-off scores as evidence of skill; shoulder-load feedback for weaver health
Puppetry	Intangible heritage scheme through Sangeet Natak Akademi [26]; artists in schools on bagless days [55]; artists-in-residence in universities [28]	Recorded, credited fingerprints of puppeteers; a teaching kit for workshops
All	National audiovisual archive, a certified trusted repository [50]	A new kind of record: how masters move

- **Cost:** about ₹19,000–31,000 for the two-sleeve prototype, falling with a custom board; a whole kit costs less than four months of a guru's state honorarium [27] and a small fraction of a motion-capture suit [65].
- **Who pays:** institutions buy kits and a yearly licence; consenting masters receive a share whenever their fingerprints are used; grants and CSR support preservation. **These are hypotheses: our first-year test is three signed letters of intent.**
- **Rules:** consent is explicit and recorded; data stays on the device or the user's app by default, in line with India's data protection law [34]; performers' rights are respected [35].

10. Impact

Who	Benefit	How we will measure it
Masters	Their own way of playing, weaving or puppeteering preserved, credited and shared on their terms	Masters recorded; their feedback
Learners	Practise against a real master between lessons, and see honest progress	Unaided score after a break
Weavers	Feedback on shoulder posture during long weaving hours [69]	Time with the shoulder raised, before and after (to be tested)
Archives	A record of how masters move, not only how they sound and look	Consented fingerprints deposited
Researchers	A consented dataset linking masters' fingerprints with learners' progress in three crafts	Dataset and publications

First-year targets [Target]

Measure	Target	Source of the number
Masters recorded with consent	15 (tabla, handloom, puppetry)	Recording register
Learners completing a two-week programme	100	App records
Unaided score vs video-only practice	Better, with confidence interval	Learning trial

Measure	Target	Source of the number
Institutions piloting	3, at least one per craft	Signed letters of intent

Possible wider benefit, not yet tested: deaf people can follow a beat through vibration [9], and India has about 50.7 lakh people with hearing disability [31]. A future study could open rhythm learning to them.

11. The pitch: what the jury will see and hear

11.1 Six things the jury should remember

1. A master's skill lives in the fingers, wrist, elbow and shoulder.
2. We measure it with one low-cost sleeve, in tabla, handloom and puppetry.
3. We prove the fingerprint is the master's, not the instrument's.
4. We teach it through touch, sound and a ghost arm.
5. We remove the help and count only what the learner does alone.
6. The master owns and controls the recorded legacy.

11.2 Three-minute demo

Time	What happens
0:00–0:20	'Archives keep how masters sound and look. Nothing keeps how their hands move.'
0:20–0:55	Two tabla masters' fingerprints on screen, and the E1 verdict: the master is still recognised on a new day and on a swapped tabla.
0:55–1:55	The judge wears the sleeve and plays along. Finger and wrist cues fade. A check cycle runs with the device silent; the unaided score appears. 'The device succeeds only when it is no longer needed.'
1:55–2:25	The judge moves a Kathputli puppet and feels which string-finger to pull. A weaver's recording replays on the ghost arm with the cloth evenness map.
2:25–2:45	The master's consent: approved with a phone fingerprint; an edited copy is rejected.
2:45–3:00	E0–E5 results, and the plan: institutions first, one platform for many crafts.

11.3 Hard questions

Question	Answer
Are you identifying the master, or the instrument, tuning, session or sensor placement?	We tested exactly that. Each master was recorded on two days with the sleeve re-fitted, and on two instruments. A fingerprint counts only if it names the master on a new day and on a new instrument, and if the same master on another instrument is closer than another master on the same instrument (Section 5).
Three crafts: isn't that spreading thin?	One sleeve, one pipeline. Tabla is validated in full; handloom and puppetry are tested on the two riskiest questions, E0 and E1. That shows the platform generalises.
Doesn't wearing sensors change how a master plays?	Fingertips and palm are free, and E0 tests it: the same tool sensor measures performance with and without the sleeve.

Question	Answer
Is there any learning evidence?	E5: beginners randomised to video or PARAMPARA, tested 48 hours later with the device off, reported with a confidence interval. It is a pilot; the full trial follows.
Why not YouTube?	Video shows the master. We measure the learner against that master and remove help until they perform alone.
Why not a motion-capture suit?	Suits cost US\$5,000 or more and do not teach [64, 65]. Our sleeve is built to teach, at a fraction of the cost.
Can vibration carry tiny timing differences?	No, and we don't ask it to. Touch carries which finger, which joint and how strong; the ear carries fine timing [43, 46].
Won't learners depend on the sleeve?	Cues fade, and only the device-off score counts [14].
Can we meet the masters?	They are named in the deck with their signed consent. With their agreement we invite them to the demo or share their contact through our institute.
Does it replace the master?	No. It extends the master's reach between lessons, under the master's control.
What if E1 fails?	We drop the style claim for that craft, keep teaching structure and rhythm, and say so.

12. Limits we state openly

- **Body sensors have limits:** shoulder angles can be off by many degrees [62]; we rely on timing, order and speed, and E1 tests the result.
- **Not captured:** grip force inside the hand, tone colour of a stroke, thread tension.
- **Touch has limits:** eight sites per arm and two or three strength levels; fast passages are taught by ear and screen.
- **Small first studies:** E0–E5 test feasibility and estimate effects; they do not prove long-term learning.
- **A craft's fingerprint may fail E1.** If so, its Style level waits.
- **Looms are not portable;** handloom work happens at weaving centres, and the finale shows recordings.
- **Copying cannot be fully prevented;** consent and credit are recorded, and licences are contractual.
- **Some masters may not want to be recorded.** Their choice decides what we do.

13. Roadmap after SIH

When	What	Starts only if
0–6 months	Tabla learning trial with about 120 people: video, video with metronome, PARAMPARA with fixed fading, PARAMPARA with check-cycle fading [78]	E0–E5 pass for tabla
0–6 months	Handloom training pilot at a Weavers' Service Centre, including shoulder-posture feedback	E0 and E1 pass for handloom

When	What	Starts only if
0–6 months	Custom flexible circuit; lower cost; stroke recognition on the device [16]	Prototype stable
6–12 months	Style level where E1 passes; archive pilot with NCAA [50]; 'Feel the Master' exhibition kiosk	Clear fingerprints in E1
12+ months	Glove and shadow puppetry, kathak footwork, pottery [21, 72]	Three-craft deployment working

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Status at the end of each entry: **V** verified by live web search in October 2026; **P** existence verified, some details from memory (check before quoting); **S** standard textbook or classic paper. Only references cited in this document are listed.

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- [67] Ministry of Textiles, Rajya Sabha answers: training for handloom workers runs through 28 Weavers' Service Centres; skill upgradation under SAMARTH reached 56,934 weavers from 2015–16 to 2020–21. **Status: P.** https://rsdebate.nic.in/bitstream/123456789/719227/1/PQ_254_22072021_U441_p355_p355.pdf
- [68] Textiles Committee. Qualification 'Handloom Weaving – Two Treadles' (TC HLM 06, NSQF Level 3): 315 hours of hands-on training, covering motions of weaving, two-treadle weaving and quality maintenance. **Status: V.** https://textilescommittee.nic.in/sites/default/files/2021-07/Handloom_Weaving_Two_Treadles.pdf
- [69] Siddiqui, L. A., Banerjee, A., Chokhandre, P., & Unisa, S. (2021). Prevalence and predictors of musculoskeletal disorders (MSDs) among weavers of Varanasi, India: a cross-sectional study. *Clinical Epidemiology and Global Health*. $n = 364$; 12-month prevalence in handloom vs powerloom weavers: upper back 84% vs 45%, lower back 82% vs 50%, shoulder 76% vs 42%. **Status: V.** https://oalib-perpustakaan.upi.edu/Record/doaj_18bf152015c74b8aae5783c503ac2069
- [70] Automatic inspection of woven fabric density based on digital image analysis (wavelet-based image method). Relative error below 0.86% compared with manual counting across three basic weaves. **Status: P.** <https://global-sci.com/article/86550/automatic-inspection-of-woven-fabric-density-based-on-digital-image-analysis>
- [71] UNIMA World Encyclopedia of Puppetry Arts, 'Kathputli': strings are looped around the puppeteer's hand and fingers, with no control bar; usually two to five strings (the rasdhari dancer up to nine). Bhat and Nat communities, mostly from Nagaur. **Status: V.** <https://wepa.unima.org/en/?p=3094>

- [72] India's four puppet families: string (Kathputli, Kundhei, Gombeyatta, Bommalattam), shadow (Tholu Bommalata, Togalu Gombeyatta, Ravanachhaya), rod (Putul Nautch, Yampuri) and glove (Pavakoothu of Kerala). **Status: V.** <https://compass.rauias.com/art-culture/puppet-forms-india/>
- [73] People's Archive of Rural India (PARI), 'Delhi's kathputli artists: puppeteers in limbo', and press reports: after the 2017 demolition of the Kathputli Colony at Shadipur, about 2,800 families of puppeteers and other folk artists were to move to a transit camp at Anand Parbat. **Status: V.** <https://ruralindiaonline.org/en/articles/delhis-kathputli-artists-puppeteers-in-limbo>
- [74] CRAEFT: Craft Understanding, Education, Training, and Preservation for Posterity and Prosperity. EU-funded project from 1 March 2023, three years, budget €3,112,106; digital aids, craft simulators and haptics for craft training. **Status: V.** https://www.isti.cnr.it/en/research/project-detail/14501/CRAEFT_Craft_Understanding_Education_Training_and_Preservation_f_or_Posterity_and_Prosperty
- [75] Mingei (EU Horizon 2020): silk-weaving pilot at the Haus der Seidenkultur, Krefeld; motion capture of craftspeople's gestures and tool use for education, AR/VR and virtual characters. **Status: V.** <https://waag.org/en/project/mingei>

P. Validation methods

- [76] Flexer, A., & Schnitzer, D. (2010). Effects of album and artist filters in audio similarity computed for very large music databases. Computer Music Journal. The 'album effect': artist identification looks accurate when training and test songs share an album, and drops when they do not, because systems learn recording conditions as well as the artist. **Status: V.**
- [77] Lakens, D. (2017). Equivalence tests: a practical primer for t tests, correlations, and meta-analyses. Social Psychological and Personality Science. Two one-sided tests (TOST) to show that an effect is smaller than a pre-set bound. **Status: V.** <https://pmc.ncbi.nlm.nih.gov/articles/PMC5502906/>

Q. Research methods

- [78] Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Erlbaum. **Status: S.**

Appendix A. Partner kit for three crafts

A.1 Whom to approach

Craft	Where to find masters
Tabla	Gurus empanelled with a Zonal Cultural Centre under Guru-Shishya Parampara [27]; tabla faculty at music colleges; Gandharva Mahavidyalaya centres [30]
Handloom	Weavers' Service Centres [67]; master weavers and trainers of the two-treadle qualification [68]; nearby handloom clusters
Puppetry	Kathputli families and troupes, including the former Kathputli Colony community [73]; puppeteers listed by Zonal Cultural Centres and Sangeet Natak Akademi [26]

A.2 Outreach message

Respected Guruji / Ustad, we are engineering students building PARAMPARA for Smart India Hackathon 2026. It records how a master moves while playing, weaving or working a puppet (light sensors on the back of the hand and arm, nothing on the fingertips or palm), so that learners can practise against it, with the master's consent and credit. We would be grateful for two short sessions of about 45 minutes on different days. You decide how the recording may be used, you will receive a copy, and you can withdraw it at any time.

Pranam Guruji / Ustad ji, hum engineering students hain aur Smart India Hackathon 2026 ke liye PARAMPARA bana rahe hain. Isme guru ke haath aur baanh ki gati record hoti hai (haath ke peeche aur baanh par halke sensor, ungliyon ke siron aur hatheli par kuch nahi), taaki shishya uske saath abhyas kar sakein, guru ki anumati aur naam ke saath. Kya aap alag-alag din do baar lagbhag 45 minute de sakte hain? Recording ka upyog aap tay karenge, aapko copy milegi, aur aap kabhi bhi wapas le sakte hain.

A.3 Consent form (plain language)

I agree that...	Yes / No
my movements may be recorded with sensors on the back of my hands and arms, and a sensor on my instrument, loom or puppet	☐ / ☐
I may be asked to play, weave or perform on a second instrument, loom or puppet	☐ / ☐
the recording may be used to teach learners through PARAMPARA	☐ / ☐
my name, gharana or community may be shown with the recording	☐ / ☐
the recording may be shown at Smart India Hackathon 2026	☐ / ☐
the recording may be used commercially (only with a separate written agreement)	☐ / ☐
a person I name may manage the recording after me: _____	☐ / ☐
I understand I can withdraw at any time, and I will receive a copy of my recording	☐

Signature, date and witness. A spoken consent recording in the master's language is kept with the form. Institute ethics approval is obtained before any study with volunteers (E2–E5).

A.4 What the master receives

- A copy of their recording and fingerprint.
- Credit wherever it is used.
- Control: approve, limit or withdraw.
- A share of any future licence income, as agreed in writing.

Appendix B. Experiment E1 protocol: is the fingerprint the master's?

B.1 Recording matrix (per craft)

	Day 1	Day 2 (sleeve re-fitted)	Cycles
Master A	Instrument 1, then 2	Instrument 2, then 1	$4 \times 20 = 80$
Master B	Instrument 2, then 1	Instrument 1, then 2	80
Master C	Instrument 1, then 2	Instrument 2, then 1	80

Instruments: two tabla sets, two looms at one weaving centre, or two puppets of the same type. The order is counterbalanced so fatigue and warm-up do not line up with an instrument.

B.2 Procedure

1. **Set-up:** sleeves on both arms, 20-second N-pose; tool sensor fitted; tuning, loom settings or puppet noted.
2. **Warm-up:** 2 minutes of free performance.
3. **Recording:** at least 20 uninterrupted cycles per instrument: Teentaal theka at 80 beats per minute (tabla); 20 picks of plain weave (handloom); a fixed gesture phrase chosen by the master, repeated 20 times (puppetry).
4. **E0 block:** on day 1, add alternating blocks of 10 cycles with and without the sleeve on instrument 1.
5. **Export** one row per cycle to a CSV file with the columns master, session, instrument, f1 ... fn (the cycle features from Section 4.3; the export is part of the week-3 build).
6. **Analyse:** `python3 PARAMPARA/sim/skilltwin_validation.py analyze e1.csv`.
7. **Report** T-a, T-b, T-c, the control, chance levels and the verdict, plus each master's profile with error bars.

Result template for the deck

Craft ____ · masters A, B, C · new day ____% · new instrument ____% (chance 33%) · confound gap ____,
p = ____ · instrument control ____% (chance 50%) · verdict ____.

Appendix C. Experiments E0 and E2–E5 protocols

#	Procedure	Measure	Pass [Target]
E0	Alternating blocks (no sleeve, sleeve, no sleeve, sleeve), 10 cycles each; tool sensor on throughout; master rates comfort	Difference in timing, cycle time and strength; equivalence by two one-sided tests [77]	Within ± 10 ms, $\pm 5\%$, ± 1 dB; comfort ≥ 4 of 5
E2	Find the weakest cue each person can feel (up-down method), at rest and while performing slowly; finger, wrist, elbow and shoulder sites; cue before vs on the beat	Ratio of 'while performing' to 'at rest' strength	≤ 2 at one or more sites per joint
E3	Feel and hear A (master 1), B (master 2), then X (one of them, at random). Answer 'A or B'. 20 rounds	Correct answers out of 20	≥ 15 of 20 ($p = 0.02$)
E4	While performing, 40 random cues across 5 fingers and 3 joints at 2 strengths; the person calls out what they felt	Percentage correct	$\geq 90\%$ sites, $\geq 80\%$ strengths
E5	16 beginners randomised (8 and 8) to video practice or PARAMPARA; three 20-minute sessions on three days on the tabla theka; retention test 48 hours later with the device off; the tester does not know the group	Unaided score at retention (primary); score during practice (secondary)	Difference and 95% CI reported; full trial if PARAMPARA $\geq$ video

All are short, low-risk tasks, seated, with comfortable vibration levels and the right to stop at any time. Written consent and the institute's ethics process apply.

Appendix D. Demo day checklist

- Two sleeves plus one spare; finger rings in three sizes; spare batteries, power bank and cables.
- Tabla and practice pad; a Kathputli puppet with its sensor; handloom recordings and a cloth sample.
- Pre-calibrated judge profile with a 20-second N-pose re-check.
- Offline copy of the app on the demo laptop.
- Masters' fingerprints and E1 verdicts loaded; one signed recording and one edited copy.
- Printed one-page summary and the masters' consent summaries.
- Backup video of the full demo; three rehearsals completed without help.

Appendix E. Simulations (planning tools, not evidence of learning)

E.1 How much to record for E1

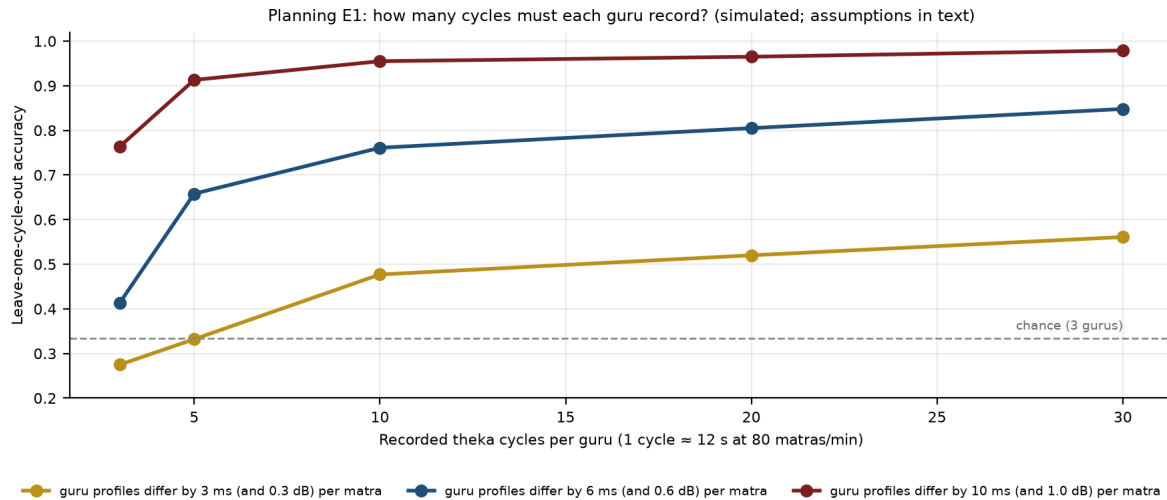


Figure E1. Simulation with assumed variation between tabla cycles (15 ms, 1.5 dB). With 20 cycles per player, players whose profiles differ by about 6 ms and 0.6 dB per beat are told apart about 80% of the time; at 10 ms, about 96%.

This sets the recording length (**at least 20 cycles per master per condition**). The real difference between masters is unknown until E1.

E.2 The confound test on synthetic data

Test (synthetic, 3 masters × 2 days × 2 instruments × 20 cycles)	World 1: master-driven	World 2: instrument-driven
Naive random split	96%	61%
T-a New day	85%	29%
T-b New instrument	96%	52%
T-c Confound gap (p)	+0.77 (p = 0.002)	−1.92
Control: name the instrument	66%	100%
Verdict	PASS	FAIL

World 2 shows why the naive test is dangerous: it looks well above chance although the 'fingerprint' is really the instrument. Code and results: `PARAMPARA/sim/skilltwin_validation.py`, `skilltwin_validation_demo.json`.

E.3 Does fading by check cycles help?

We simulated 1,000 learners over three practice days with overnight forgetting and very different learning speeds, under three assumptions about how guidance affects learning.

Unaided score on day 4 (simulated)	Assumption A	Assumption B	Assumption C
Cues always on	0.48	0.42	0.57
No cues	0.46	0.58	0.71

Unaided score on day 4 (simulated)	Assumption A	Assumption B	Assumption C
Fixed fade	0.67	0.63	0.68
Check-cycle fade (PARAMPARA)	0.69	0.65	0.68

Cues that never stop do poorly under every assumption. Check-cycle fading did slightly better than a fixed fade under two assumptions (+0.015, 95% CI 0.013–0.017) and equally under the third. The gain is small and depends on the assumptions, so people, not simulations, decide (E5 and the full trial). Code: PARAMPARA/sim/fade_engine_sim_v2.py.

Appendix F. Reference check and parameters

F.1 References removed from the original team documents

Original citation	Finding
'Masur & Sacks, Gesture-Based Adaptive Haptic Guidance, IEEE Robotics (2025)'	Does not exist; closest real work is Zahedi et al. 2017 [11]
'Flandorfer et al., Wearable Haptic Learning Systems, Sensors (2022)'	Not found; replaced by a verified review [10]
'Grindlay (2008), ACM Multimedia'	Wrong venue; correct sources cited [1, 2]
Chat-tool markers such as 【35†L1-L4】	Not references; removed

F.2 Starting parameters

Parameter	Value
Motion sensor sampling	100–200 samples per second
Tabla stroke sensor sampling	≥ 4,000 samples per second
Fingerprint recording	≥ 20 cycles per master, day and instrument
Calibration	20-second N-pose each session
Cue length	30–80 ms
Cue strength	≥ 3 × the person's weakest felt level
Cue ends before movement	about 100 ms (measured per person)
Check cycles	session start + every 4 cycles
Target to move on	unaided score ≥ 85% on 2 days + master's approval
E1 pass	T-a and T-b ≥ 70%; T-c p < 0.01